

GEOLOGY FINAL WEAPON

70-Mark Exam · Target: 65+ · 564 Rudra Singh XI-C

Paper Pattern: Q1A MCQ (10) | Q1B 1-word (8) | Q2 SA-2mk (16) | Q3 SA-3mk (24) | Q4 LA (12)

Portion: Petrology · Wind · River · Glacier · Sea (Depositional) · Mineralogy · Intro to Geology · Bedding & Superposition

STRATEGY: Section 1 (MCQ/1-word bank) is your weakness target grind it first and last. Sections 2–8 cover the portion in textbook order. Section 9 is the 5-minute pre-exam revision board.

1. THE GOLDEN 18 MARKS: MCQ & 1-WORD BANK

Intro to Geology

- Study of Earth = **Geology** (Geo + Logos)
- Father of Geology = **James Hutton**
- “Present is key to past” = **Uniformitarianism**
- Outermost solid layer = **Crust**
- Crust + upper mantle = **Lithosphere**
- Semi-fluid layer below lithosphere = **Asthenosphere**
- Crust–mantle boundary = **Moho discontinuity**
- Mantle–core boundary = **Gutenberg discontinuity**
- Outer core is **liquid Fe–Ni**; inner core is **solid Fe–Ni**
- Study of minerals = **Mineralogy**
- Study of rocks = **Petrology**
- Study of fossils = **Palaeontology**
- Study of rock layers = **Stratigraphy**
- Study of landforms = **Geomorphology**

Weathering & Surface

- Freeze–thaw effect on rock = **frost wedging**
- Angular fragments at cliff base = **scree/talus**
- Layers peeling from granite dome = **exfoliation**
- Most stable mineral = **quartz**
- Least stable mineral = **olivine**
- Mechanical weathering dominates in = **hot & arid**
- Chemical weathering dominates in = **hot & humid**
- Hematite → Limonite process = **hydration**
- Feldspar → clay process = **hydrolysis**
- $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ = **oxidation**
- Spheroidal weathering seen in = **basalt**
- Central unweathered mass = **core stone/residual boulder**
- Most productive soil layer = **A horizon**
- Zone of accumulation = **B horizon**
- Study of soil = **Pedology**

Wind

- Most active agent in desert = **Wind**
- Removal of loose material by wind = **deflation**
- Depression from deflation = **blowout**
- Wind-polished/faceted rock = **ventifact**
- 3-faced ventifact = **dreikanter**
- Elongated wind-carved ridge = **yardang**
- Main wind transport mode = **saltation**
- Fine wind-deposited silt = **loess**
- Crescent-shaped dune = **barchan**
- Barchan horns point = **downwind**
- Parabolic dune horns point = **upwind**
- Long ridge parallel to wind = **seif/longitudinal**
- Pyramidal multi-armed dune = **star dune**

River

- Force of water dislodging rock = **hydraulic action**
- S-shaped river bend = **meander**
- Cut-off meander = **oxbow lake**
- Circular bed depression = **pothole**
- Fine material in turbulent water = **suspended load**
- Rolling + bouncing of pebbles = **bed load** (traction + saltation)
- Cone at mountain base = **alluvial fan**
- Raised river bank = **natural levee**
- Triangular river-mouth deposit = **delta**
- Nile delta type = **arcuate**
- Mississippi delta type = **bird's foot/digitate**
- Tiber delta type = **cusped**
- Inside bend of meander deposit = **point bar**

Glacier

- Slow-moving mass of ice = **glacier**
- Antarctica / Greenland type = **continental/ice sheet**
- Mountain valley glacier = **valley/alpine**
- Bowl at glacier head = **cirque/corrie**
- Knife-edge ridge between cirques = **arete**
- Pyramidal peak (Matterhorn) = **horn**
- Glacial valley cross-section = **U-shaped**
- Drowned U-valley on coast = **fjord**
- Tributary valley above main valley = **hanging valley**
- Asymmetric glacial bedrock hill = **roche moutonnée**
- Unsorted glacial debris = **till/moraine**
- Middle moraine of 2 merged glaciers = **medial moraine**
- Foreign boulder from glacier = **erratic**
- Oval till hill in swarms = **drumlin**
- Winding meltwater ridge = **esker**
- Depression from melted ice block = **kettle**

Sea (Depositional)

- Sand/pebble shore deposit = **beach**
- Ridge from shore, one end free = **spit**
- Ridge across bay mouth = **bar**
- Bar connecting island to mainland = **tombolo**
- Shallow water behind bar = **lagoon**
- Long sandy island parallel to coast = **barrier island**
- Transport along coast by waves = **longshore drift**

Petrology & Mineralogy

- Granite & rhyolite have same = **composition**
- Rocks rich in mafic minerals = **melanocratic**
- Thermal metamorphism of sandstone = **quartzite**
- Thermal metamorphism of limestone = **marble**
- Cataclastic of argillaceous = **slate**
- Discordant igneous bodies = **dyke, batholith**
- Concordant igneous body = **sill**
- Rock with highest metamorphism = **gneiss**
- $>66\% \text{ SiO}_2$ = **acidic**
- $<44\% \text{ SiO}_2$ = **ultrabasic**
- Silicates % of crust = **$>95\%$**
- Mohs scale is = **arbitrary, non-linear**
- Softest mineral = **talca (1)**
- Hardest mineral = **diamond (10)**
- Most abundant crustal mineral group = **feldspar (50%)**
- 2nd most abundant mineral = **quartz**
- Amphibole cleavage angle = **124°**
- Pyroxene cleavage angle = **$\sim 90^\circ$**
- Amphibole chain type = **double chain inosilicate**
- Pyroxene chain type = **single chain inosilicate**
- Mica structure = **phyllosilicate**
- Quartz structure = **tectosilicate**
- Olivine structure = **nesosilicate**
- Beryl structure = **cyclosilicate**
- Epidote structure = **sorosilicate**
- “Fool's gold” = **pyrite**
- Iceland spar variety of = **calcite (double refraction)**
- Effervesces with dilute HCl = **calcite**
- Father of Mineralogy = **Agricola Georgius**
- First mineral named after a person = **Prehnite**

Bedding & Stratigraphic Principles

- Oldest at bottom = **Law of Superposition (Steno)**
- Originally flat layers = **Original Horizontality**

- Layers extend laterally = **Lateral Continuity**
- Cross-cutting feature is = **younger**
- Fossils in definite order = **Faunal Succession (W. Smith)**
- Tilted below, horizontal above = **angular unconformity**
- Sedimentary on igneous basement = **nonconformity**
- Parallel beds with erosion gap = **disconformity**
- Coarse bottom, fine top = **graded bedding**
- Inclined layers within a bed = **cross-bedding**
- Polygonal dried cracks = **mud cracks**
- Short-lived, widely distributed fossil = **index fossil**
- Rock containing inclusions is = **younger**

2. INTRODUCTION TO GEOLOGY

- **Geology** = study of Earth's origin, structure, composition, history. Greek *Geo* (Earth) + *Logos* (science).
- **Branches:** Petrology, Mineralogy, Stratigraphy, Palaeontology, Structural Geology, Geomorphology, Hydrogeology, Economic Geology.
- **Earth's interior (in order):** Crust → Mantle → Outer core → Inner core.
 - *Crust:* Continental (granitic/sialic, 35–70 km) + Oceanic (basaltic/simatic, 5–10 km).
 - *Mantle:* Upper + Lower, to 2900 km. Asthenosphere (semi-molten) lies in upper mantle.
 - *Core:* 2900–6371 km. Outer = liquid Fe-Ni (generates magnetic field). Inner = solid Fe-Ni. Temperature ~5000°C.
- **Lithosphere** = crust + rigid upper mantle. Broken into tectonic plates.
- **Moho discontinuity** = crust–mantle boundary. **Gutenberg discontinuity** = mantle–core boundary (2900 km).
- **Uniformitarianism (Hutton):** “Present is the key to the past.”

3. BEDDING & STRATIGRAPHIC PRINCIPLES (YOUR WEAKNESS FIXED)

3.1 Bedding & Stratification

Bed/Stratum = layer of sedimentary rock with distinct composition/grain size/colour. **Bedding plane** = surface separating two successive beds, marks change in deposition.

Types of bedding (sedimentary structures):

- **Horizontal bedding** original undisturbed flat layers.
- **Cross-bedding (current bedding)** inclined layers within a main bed; formed by flowing water/wind; indicates current direction.
- **Graded bedding** coarse particles at bottom, fine at top; formed by turbidity currents; indicates way-up (which side is up).
- **Ripple marks** small undulations on bedding surface from waves/current.
- **Mud cracks** polygonal cracks from drying of wet mud; indicates subaerial exposure.

3.2 Steno's Three Laws (1669)

1. **Law of Superposition:** In an undisturbed sequence, the **oldest bed is at the bottom, youngest at the top**. Basis for relative dating.
2. **Law of Original Horizontality:** Sedimentary layers are deposited originally horizontally under gravity. Any tilting/folding occurred later.
3. **Law of Lateral Continuity:** Sedimentary layers extend laterally in all directions until they thin out or meet a barrier.

3.3 Other Principles

- **Cross-cutting Relationships (Hutton):** A feature (fault, intrusion) that cuts another is **younger** than the one it cuts.
- **Faunal Succession (William Smith):** Fossil organisms succeed each other in a definite order. Rocks of same age contain same fossils.
- **Inclusions:** A rock containing inclusions is **younger** than the rock fragments it contains.

3.4 Unconformities (Gaps in Geological Record)

- **Angular unconformity** older beds tilted/folded, younger beds horizontal above.
- **Disconformity** both beds parallel, but erosion surface between.
- **Nonconformity** sedimentary beds over igneous/metamorphic basement.
- **Paraconformity** beds parallel, no obvious erosion, only fossil gap.

RELATIVE vs ABSOLUTE dating: Relative = order only (superposition, cross-cutting, fauna). Absolute = actual age in years (radiometric: C-14, U-Pb, K-Ar).

4. PETROLOGY

4.1 Rock Types (3 classes)

Petrology = study of rocks. **Petrography** = rocks under microscope. **Petrogenesis** = origin of rocks. **Lithology** = hand-specimen description. **Exogenic** = Sun-driven surface processes (weathering, sedimentary). **Endogenic** = internal heat-driven (volcanism, metamorphism, igneous).

4.2 Igneous Rocks

Magma: 800–1400°C. Volatile (H₂O, CO₂, SO₂) + non-volatile (oxides of Si, Al, Fe, Ca, Na, K, Mg).

SiO ₂ %	Type	Plutonic	Hypabyssal	Volcanic
>66%	Acidic	Granite	Pegmatite	Rhyolite
65–55%	Intermediate	Syenite	–	–
54–44%	Basic	Gabbro	Dolerite	Basalt
<44%	Ultrabasic	Dunite	–	–

- **Plutonic** deep, slow cooling, large crystals, equigranular, holocrystalline (e.g. granite).
- **Hypabyssal** medium depth, moderate cooling, mixed grains, porphyritic (e.g. dolerite).
- **Volcanic** surface, rapid cooling, fine/glassy, holohyaline (e.g. basalt).
- **Felsic** = light (Si+Al), feldspar + quartz. **Mafic** = dark (Fe+Mg), olivine/pyroxene/amphibole.

9 characteristics of igneous rocks: primary rocks, massive/unstratified, non-fossiliferous, occur as veins/fissures, baking at contacts, interlocking texture, gas cavities/amygdales in lava, pyrogenetic minerals, equigranular/porphyritic/glassy texture.

4.3 Sedimentary Rocks

Weathering → erosion → transport → deposition → **lithification**. **Clastic** = broken fragments. **Non-clastic** = chemical precipitation. **Bioclastic** = organic.

Class	Grain size	Examples
Rudaceous	256–2 mm (boulders, pebbles)	Conglomerate, Breccia
Arenaceous	2–0.06 mm (sand)	Sandstone
Argillaceous	<0.06 mm (clay, mud)	Shale, Mudstone

6 characteristics: layered/stratified, clastic textures, may contain fossils, primary structures (ripple marks, mud cracks), tracks/trails, derived from pre-existing rocks.

4.4 Metamorphic Rocks

Agents: temperature (most important), pressure (directional + uniform), chemically active fluids.

Type	Agent	Parent → Product	Structure
Cataclastic	Directional pressure	Argillaceous → Slate	Slaty cleavage
Thermal	Temperature	Sandstone → Quartzite; Limestone → Marble	Granulose
Dynamothermal	Heat + Pressure	Shale/Basalt → Schist/Gneiss	Schistose, Gneissose

Characteristics: from pre-existing rocks, solid-state transformation, recrystallization, interlocking minerals, schistosity/gneissosity, characteristic minerals (kyanite, garnet, sillimanite, staurolite), **no fossils**.

4.5 Rock Cycle

Igneous $\xrightarrow{\text{weathering}}$ Sedimentary $\xrightarrow{\text{heat+pressure}}$ Metamorphic $\xrightarrow{\text{melting}}$ Magma $\xrightarrow{\text{cooling}}$ Igneous. Any rock can become any other type. Driven by exogenic + endogenic forces.

5. MINERALOGY

5.1 5 Criteria for a Mineral

(1) Natural, (2) Inorganic, (3) Solid, (4) Definite chemical composition, (5) Ordered internal atomic structure. **NOT minerals:** petroleum, water, mercury (liquid), pearls/coral (organic), cubic zirconia (synthetic), natural glass (no structure).

5.2 Physical Properties

Mohs scale (arbitrary, non-linear): 1-Talc 2-Gypsum 3-Calcite 4-Fluorite 5-Apatite 6-Orthoclase 7-Quartz 8-Topaz 9-Corundum 10-Diamond. Soft <2 (fingernail), Medium 3–5 (knife), Hard 6–9 (scratches glass), H>7 = gemstone grade.

Lustre types: Metallic (pyrite), sub-metallic (hematite), adamantine (diamond), vitreous (quartz most common), pearly (talc/mica), silky (satin spar gypsum), resinous (amber), waxy (chalcedony), dull/earthy (kaolinite, bauxite).

Streak (more reliable than colour): Pyrite = golden but grey streak. Hematite = dark but **cherry/brick red streak**.

Fracture: Conchoidal (obsidian), earthy (bauxite), hackly (copper sharp projections), splintery (kyanite), uneven (magnetite).

Cleavage (sets + angles): 1 set = mica. 2 at 90° = feldspar. 2 at 124° = amphibole. 3 at 90° = galena. 3 at 105°/75° = calcite. 4 sets = fluorite (octahedral).

5.3 Silicate Classification

Structure	Arrangement	Example
Nesosilicate	Isolated tetrahedra	Olivine, Garnet, Kyanite
Sorosilicate	Pairs	Epidote
Inosilicate (single chain)	Single chain	Pyroxene (Augite)
Inosilicate (double chain)	Double chain	Amphibole (Hornblende)
Cyclosilicate	Rings	Beryl, Tourmaline
Phyllosilicate	Sheets	Mica, Talc, Serpentine
Tectosilicate	3D framework	Quartz, Feldspar, Zeolite

5.4 Key Mineral Groups (Quick Reference)

Mineral	Structure	H	Cleavage	Key feature
Feldspar	Tectosilicate	6–6.5	2 sets, 90°	50% of crust. Plagioclase series: Ab-Ol-An-La-By-An
Quartz	Tectosilicate	7	None	2nd most abundant. Conchoidal fracture. SiO ₂
Amphibole	Inosilicate (double)	5–6	2 at 124°	Dark. HAS OH. Hornblende is common
Pyroxene	Inosilicate (single)	5–6	2 at 90°	Dark. NO OH. Augite common
Mica	Phyllosilicate	2–3	1 set, perfect	Thin elastic plates. Muscovite / Biotite
Olivine	Nesosilicate	6.5–7	None	Olive green. Forsterite(Mg) – Fayalite(Fe)
Zeolite	Tectosilicate	3.5–5	Perfect 1–2	Basalt cavities. “Molecular sieves”
Calcite	Carbonate	3	3 sets, perfect	CaCO ₃ . Effervesces with HCl. Iceland spar
Gypsum	Sulphate	2	2 sets	CaSO ₄ ·2H ₂ O. Plaster of Paris
Fluorite	Halide	4	4 sets (octa)	CaF ₂ . Deeply coloured
Talc	Phyllosilicate	1	1 set	Softest. Soapstone

6. GEOLOGICAL WORK OF WIND (AEOLIAN)

Wind is the most active erosion/deposition agent in arid and semi-arid regions.

6.1 Erosion Processes

Deflation blowing away loose fine material; creates deflation hollows/blowouts. **Abrasion (corrasion)** wind-driven sand sandblasts rocks. **Attrition** sand grains collide, become smaller and rounder.

6.2 Erosional Landforms

- **Ventifact** rock polished/faceted by wind-driven sand. 3 faces = *dreikanter*.
- **Yardang** elongated ridge parallel to wind direction; alternating hard/soft rock.
- **Mushroom/Pedestal rock** narrow base (sand near ground), broad top.
- **Inselberg** isolated residual steep-sided hill (resistant rock) on a plain.
- **Zeugen** table rock with hard cap on soft base.
- **Deflation hollow** depression; reaches water table = oasis.

6.3 Transport Modes

Suspension fine dust, long distance. **Saltation** sand grains bounce (main mode). **Surface creep (traction)** large particles roll.

6.4 Depositional Landforms

- **Barchan** crescent-shaped, isolated, **horns point downwind**. Most common.
- **Parabolic** U-shaped, **horns point upwind** (opposite of barchan!). Partially vegetated.
- **Seif (longitudinal)** long narrow ridge parallel to wind.
- **Transverse** ridge perpendicular to wind; abundant sand supply.
- **Star dune** multiple arms from central peak; variable wind directions.
- **Loess** fine wind-deposited silt. Very fertile. China, Europe.

TRAP: Barchan horns = DOWNWIND. Parabolic horns = UPWIND. They are visual mirror images.

7. GEOLOGICAL WORK OF RIVER (FLUVIAL)

7.1 Erosion Processes (4)

Hydraulic action (force of water), **Abrasion/corrasion** (sediment grinds bed), **Attrition** (sediment collides & breaks), **Corrosion/solution** (chemical dissolution, esp. limestone).

7.2 Erosional Landforms

- **V-shaped valley** upper course, vertical erosion dominant.
- **Gorge/canyon** deep, narrow, near-vertical walls (e.g. Grand Canyon).
- **Waterfall** river drops over hard rock; soft rock undercut below.
- **Pothole** circular bed depression from swirling pebbles in eddies.
- **Meander** S-shaped bend. Erosion outside (cutbank, faster flow), deposition inside (point bar, slower).
- **Oxbow lake** crescent lake formed when meander is cut off from main channel.

7.3 Transport Modes

Dissolved load (invisible ions in solution), **Suspended load** (fine silt/clay/sand held by turbulence makes river muddy), **Bed load** = traction (rolling) + saltation (bouncing) of pebbles/boulders.

7.4 Depositional Landforms

- **Alluvial fan** cone at mountain base where river hits plain; coarse sediment.
- **Floodplain** flat area alongside river; built by repeated flooding.
- **Natural levee** raised bank built up by flood deposits.
- **Point bar** inner meander side deposit.
- **Braided channel** multiple channels around sediment bars; heavy load.
- **Delta** triangular deposit at river mouth.

7.5 Types of Deltas

- **Arcuate (fan-shaped)** Nile, Ganga–Brahmaputra.
- **Bird's foot (digitate)** finger-like extensions; Mississippi.
- **Cuspate** tooth/arrow-shaped; wave-shaped; Tiber (Italy).
- **Estuarine** funnel-shaped in river estuary.

8. GEOLOGICAL WORK OF GLACIER

Glacier = slowly moving mass of ice on land, formed by compaction and recrystallization of snow. **Snowline** = altitude above which snow persists year-round.

8.1 Types of Glaciers (4)

- **Continental/Ice sheet** vast thick ice (Antarctica, Greenland).
- **Valley/Alpine** tongue of ice in a mountain valley (Siachen, Gangotri).
- **Piedmont** valley glacier spreads onto flat plain at mountain base (Malaspina, Alaska).
- **Cirque** small glacier in a bowl-shaped depression on mountainside.

8.2 Erosion: Plucking + Abrasion

Plucking ice freezes onto rock, pulls fragments away as it moves. **Abrasion** embedded rocks in ice grind the bedrock, creating **striations**.

8.3 Erosional Landforms

- **Cirque/Corrie** bowl-shaped depression at glacier head.
- **Arete** knife-edge ridge between two cirques.
- **Horn** pyramidal peak where 3+ cirques erode from different sides (Matterhorn).
- **U-shaped valley** glacier widens & deepens original V-valley; flat floor, steep sides.
- **Hanging valley** tributary glacier valley left above main valley; waterfall at mouth.
- **Fjord** drowned U-valley along coast (Norway).
- **Roche moutonnée** asymmetric bedrock hill; smooth upstream (abrasion), rough downstream (plucking).

8.4 Depositional Landforms

- **Moraine (till)** unsorted glacial debris. Types: **Lateral** (sides), **Medial** (centre when 2 glaciers merge), **Terminal/end** (front), **Ground** (bottom).
- **Erratic** large boulder carried far from source by glacier, deposited on different rock type when ice melts.
- **Drumlin** elongated oval hill of till; steep upstream, gentle downstream; occurs in swarms.
- **Esker** long winding ridge of sand/gravel from meltwater stream flowing within/under glacier.
- **Outwash plain** flat sand/gravel deposited by meltwater streams beyond terminal moraine.
- **Kettle lake** depression formed when buried ice block melts, fills with water.

9. GEOLOGICAL WORK OF SEA: DEPOSITIONAL FEATURES

Agents: waves, tides, currents, **longshore drift** (transport along coast).

- **Beach** accumulation of sand/pebbles/shells along shore. Most common coastal deposit.
- **Spit** narrow ridge of sand/gravel extending from shore into open water; one end free; curved end = hook.
- **Bar** ridge of sand across a bay mouth. Fully closing = encloses a lagoon.
- **Tombolo** bar connecting an island to the mainland.
- **Barrier island** long narrow sandy island parallel to coast, separated by lagoon.
- **Lagoon** shallow body of water separated from sea by bar or barrier island.
- **Raised beach** former beach now above current sea level (uplift or sea-level drop).
- **Marine terrace** wave-cut platform now elevated.

Spit vs Bar: Spit attached at ONE end only. Bar connects two land points or closes a bay.

10. LIKELY EXAM QUESTIONS (PRACTICE LIST)

10.1 Short Answers (2–3 marks)

1. Classify igneous rocks on the basis of SiO_2 percentage (table).
2. Describe the three types of metamorphism with products.
3. Explain any 3 erosional landforms by wind.
4. Describe depositional landforms by river.
5. Describe the types of glaciers with examples.
6. What are erratics? What are moraines?
7. Types of deltas with examples.
8. Distinguish between spit and bar.
9. Explain Mohs' scale of hardness. Why is it non-linear?
10. Silicate classification with examples.
11. Difference between amphibole and pyroxene.
12. State Steno's three laws.
13. Explain types of unconformities.
14. Explain graded bedding and cross-bedding.
15. What is an index fossil? Give examples.

10.2 Long Answers (4 marks, any 3)

1. Explain the geological work of wind (erosion + deposition) in detail.
2. Explain the depositional landforms of a river with examples.
3. Describe the erosional and depositional work of glaciers.
4. Classify igneous rocks and explain each class with characteristics.
5. Describe the physical properties used in mineral identification.
6. Explain the rock cycle in detail with a flow description.
7. Discuss Steno's principles and their importance in stratigraphy.
8. Describe the silicate classification of minerals with examples.

11. FINAL 5-MINUTE REVISION BOARD

Intro: Lithosphere = crust + upper mantle. Moho = crust–mantle. Gutenberg = mantle–core. Hutton = Uniformitarianism.

Bedding: Superposition = oldest at bottom. Original horizontality = deposited flat. Cross-cutting = younger. Angular unconformity = tilted below, flat above. Nonconformity = sediment on igneous.

Petrology: Acidic $>66\%$ SiO_2 . Plutonic = slow cooling, large crystals. Volcanic = fast, fine. Cataclastic $\rightarrow$ slate. Thermal $\rightarrow$ quartzite/marble. Dynamothermal $\rightarrow$ schist/gneiss.

Mineralogy: Mohs 1–10 = Ta-Gy-Ca-Fl-Ap-Or-Qu-To-Co-Di. Amphibole = $124^\circ + \text{OH}$. Pyroxene = $90^\circ +$ no OH. Feldspar = 50% crust. Silicates: Neso (Olivine), Soro (Epidote), Ino (Py/Amp), Cyclo (Beryl), Phyllo (Mica), Tecto (Qz/Fsp).

Wind: Deflation = blowing away. Ventifact = polished. Yardang = parallel ridge. Barchan = crescent, horns DOWN. Parabolic = horns UP. Saltation = main transport.

River: Suspended = fine in water. Bed load = rolling/bouncing. Meander $\rightarrow$ oxbow lake. Deltas: Arcuate (Nile), Bird's foot (Mississippi), Cuspate (Tiber).

Glacier: Plucking + abrasion. Cirque = bowl, Arete = ridge, Horn = peak, U-valley, Fjord = drowned U. Moraine = till. Erratic = foreign boulder. Drumlin = oval hill. Esker = winding ridge.

Sea: Spit = one end free. Bar = across bay. Tombolo = island–mainland. Lagoon = behind bar.